STA2023 Exam 2 — Ch 4–6 — SIDE 1

■ BLUE = definition ■ **RED = formula / rule** ■ GREEN = example ■ BLACK = steps

SYMBOLS

Symbol	Means
P(E)	probability that event E happens (0 to 1)
N(E), N(S)	# of outcomes in E; # of outcomes in the whole sample space
E^c / "not E"	complement = everything that is NOT E
P(F | E)	probability of F **given** E already happened
${}_nC_r$, ${}_nP_r$, n!	combinations (order doesn't matter), permutations (order matters), factorial
X / x	random variable / one specific value of it
P(x)	probability that X equals x
μ_X, σ_X	mean (expected value) and standard deviation of X
n, p, x	binomial: # of trials, P(success) on one trial, # of successes
z	z-score = how many SDs a value is from the mean
Σ	"add them all up"

Round probabilities to 4 decimals. Every answer: formula → plug in → arithmetic → boxed answer.

CH 4 — PROBABILITY

Basics

- **Probability** = long-run proportion of times an outcome happens. **0 ≤ P(E) ≤ 1; all outcomes add to 1**. **Unusual** = P < 0.05.
- **Sample space S** = all possible outcomes; **event** = some of them.
- **Classical** (equally likely): **P(E) = N(E) ÷ N(S)**. **Empirical** (from data): **P(E) ≈ frequency ÷ total**. **Subjective** = educated guess.
- **Disjoint** = can't both happen (no overlap). **Independent** = one doesn't change the chance of the other.

NOT: $P(E^c) = 1 - P(E)$
OR: P(E or F) = P(E) + P(F) − P(E and F)
(disjoint → P(E and F) = 0)
AND (independent): P(E and F) = P(E)·P(F)
AND (general): P(E and F) = P(E)·P(F | E)
GIVEN: $P(F \mid E) = \frac{P(E \text{ and } F)}{P(E)} = \frac{N(E \text{ and } F)}{N(E)}$
AT LEAST ONE: P(at least 1) = 1 − P(none)
Independent test: P(F | E) = P(F)? yes → independent

Contingency table (two-way table) ★Q1

1. Add row totals and column totals; grand total = N.
2. "is A" → (A's total) ÷ N.
3. "A **or** B" → P(A) + P(B) − P(both) (subtract the overlapping cell).
4. "A **and** B" → the one cell ÷ N.
5. "A **given** B" → cell ÷ (B's row or column total). Divide by the GIVEN group only.

Q: "Employment status of 78 residents: Full-time M 22 / F 18; Part-time M 6 / F 10; Unemployed M 3 / F 2; Not in labor force M 5 / F 12. Males = 36, Females = 42."
Row totals: full-time 40, part-time 16, unemployed 5, not in labor force 17.
(a) P(male) = 36/78 = **0.4615**
(b) P(part-time) = 16/78 = **0.2051**
(c) P(male OR unemployed) = 36/78 + 5/78 − 3/78 = 38/78 = **0.4872**
(d) P(female AND not in labor force) = 12/78 = **0.1538**
(e) P(part-time | female) = 10/42 = **0.2381** (divide by females only)

Q: "P(speeder pulled over) = 0.8; P(ticket | pulled over) = 0.9. P(pulled over AND ticket)?" **A:** 0.8 × 0.9 = **0.72**
Q: "P(24-yr-old male survives the year) = 0.9986. P(at least one of 1000 dies)?" **A:** $1 - 0.9986^{1000} = 1 - 0.2464 =$ **0.7536**
Q: "One card: P(king or diamond)?" **A:** 4/52 + 13/52 − 1/52 = 16/52 = **0.3077**

Counting

Sequence of choices: multiply p·q·r…
n! = n(n−1)(n−2)⋯1, 0! = 1
Order matters: $_nP_r = \frac{n!}{(n-r)!}$ Order doesn't: $_nC_r = \frac{n!}{r!(n-r)!}$

3-letter codes: 26·26·26 = 17,576 · chair/vice/sec from 14: $14 \cdot 13 \cdot 12 = 2{,}184 \cdot {}_7P_5 = 7!/2! = 2{,}520 \cdot {}_{10}P_3 = 10 \cdot 9 \cdot 8 = 720 \cdot {}_4C_2 = 24/(2 \cdot 2) = 6 \cdot {}_{20}C_4 = (20 \cdot 19 \cdot 18 \cdot 17)/24 = 4{,}845$
P(4 cards all match) = $13 / {}_{52}C_4 = 13/270{,}725 \approx 0.00005$

CH 5 — DISCRETE RANDOM VARIABLES

Probability distribution ★Q2, Q3

- **Discrete** = countable values (0, 1, 2 …); **continuous** = any value in an interval (measured).
- **Probability distribution** = every value of X with its P(x) (table, graph, or formula).

VALID if: (1) each 0 ≤ P(x) ≤ 1 and (2) Σ P(x) = 1
$\mu_X = \Sigma [x \cdot P(x)]$ $\sigma_X = \sqrt{\Sigma [(x-\mu)^2 \cdot P(x)]}$
shortcut: $\sigma_X = \sqrt{\Sigma [x^2 \cdot P(x)] - \mu^2}$

1. P(x) = frequency ÷ total (if given counts).
2. Mean: multiply each x by P(x), add.
3. SD: (x − μ)², times P(x), add, square root.
4. Graph: probability histogram — one bar centered on each x, height = P(x).

Q: "Is this a valid discrete probability distribution? x: 0,1,2,3,4 with P(x): 0.34, 0.21, 0.13, 0.04, 0.01."
A: each P(x) is between 0 and 1 ✓, but 0.34+0.21+0.13+0.04+0.01 = 0.73 ≠ 1 → **NOT valid** (probabilities must add to 1).

Q: "Games played in the Stanley Cup, X: x = 4 (freq 20), 5 (20), 6 (26), 7 (17). (a) Construct the probability distribution. (b) Graph it. (c) Compute and interpret the mean. (d) Compute the SD."
(a) total = 20+20+26+17 = 83 → P(4) = 20/83 = 0.2410, P(5) = 0.2410, P(6) = 26/83 = 0.3133, P(7) = 17/83 = 0.2048 (sum ≈ 1 ✓)
(b) bars at x = 4, 5, 6, 7 with heights 0.2410, 0.2410, 0.3133, 0.2048
(c) μ = 4(0.2410) + 5(0.2410) + 6(0.3133) + 7(0.2048)
= 0.9640 + 1.2050 + 1.8798 + 1.4336 = **5.48 games**
→ over many series, the Cup lasts about 5.48 games on average
(d) $(4-5.48)^2(0.2410) = 2.1904(0.2410) = 0.5279$
$(5-5.48)^2(0.2410) = 0.2304(0.2410) = 0.0555$
$(6-5.48)^2(0.3133) = 0.2704(0.3133) = 0.0847$
$(7-5.48)^2(0.2048) = 2.3104(0.2048) = 0.4732$
$\sigma^2 = 1.1413 \rightarrow \sigma = \sqrt{1.1413} \approx$ **1.07 games**

Q: "Chuck-a-Luck profit: −$1 (0.5787), $1 (0.3472), $2 (0.0695), $3 (0.0046). Mean?" **A:** −0.5787 + 0.3472 + 0.1390 + 0.0138 = **−$0.08** → lose 8¢ per game on average. **Insurance:** $100(0.9986) + (−$9,900)(0.0014) = 99.86 − 13.86 = **$86** expected profit.

Binomial ★Q4

Binomial experiment = (1) fixed n trials, (2) independent, (3) two outcomes (success/failure), (4) same p every trial. X = # of successes.

$P(x) = {}_nC_x \cdot p^x \cdot (1-p)^{n-x}$ $_nC_x = \frac{n!}{x!(n-x)!}$
$\mu = n \cdot p$ $\sigma = \sqrt{n \cdot p \cdot (1-p)}$
Excel: =BINOM.DIST(x, n, p, FALSE) exactly x · TRUE = P(X ≤ x)

Words	Compute
exactly x	P(x)
fewer than k / less than k	P(0) + … + P(k − 1)
at most k / no more than k / ≤ k	P(0) + … + P(k)
more than k / > k	1 − [P(0) + … + P(k)]
at least k / ≥ k	1 − [P(0) + … + P(k − 1)]
between a and b (inclusive)	P(a) + … + P(b)

Q: "60% of U.S. women say the minimum driving age should be 18. Random sample of 10 women. (a) Exactly 3? (b) Fewer than 3? (c) Between 1 and 3 inclusive? (d) For a sample of 200, expected number and SD?"
n = 10, p = 0.60, 1 − p = 0.40
(a) $P(3) = {}_{10}C_3(0.6)^3(0.4)^7 = 120 \cdot 0.216 \cdot 0.0016384 =$ **0.0425**
(b) P(X < 3) = P(0) + P(1) + P(2)
$P(0) = (0.4)^{10} = 0.0001 \cdot P(1) = 10(0.6)(0.4)^9 = 0.0016 \cdot P(2) = 45(0.36)(0.4)^8 = 0.0106$
= 0.0001 + 0.0016 + 0.0106 = **0.0123**
(c) P(1 ≤ X ≤ 3) = P(1) + P(2) + P(3) = 0.0016 + 0.0106 + 0.0425 = **0.0547**
(d) n = 200: μ = 200(0.60) = **120**; σ = $\sqrt{200 \cdot 0.60 \cdot 0.40} = \sqrt{48} =$ **6.93**

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CH 6 — CONTINUOUS & NORMAL

Density curves & uniform ★Q5

- **Probability density function (pdf)** = a curve where the AREA under it over an interval = the probability X is in that interval; total area = 1; curve ≥ 0.
- **P(X = one exact value) = 0**, so P(a < X < b) = P(a ≤ X ≤ b).
- **Uniform on [a, b]** = flat rectangle: every interval of the same length is equally likely.

Uniform height = $\frac{1}{b-a}$ P(c ≤ X ≤ d) = $\frac{d-c}{b-a}$ = (length of piece) × height
"P% within the next t" → t = P × (b − a)

Q: "Daytime temperature X is uniformly distributed between 70°F and 100°F. (a) Sketch/describe the pdf. (b) P(75 ≤ X ≤ 85)? (c) P(X < 80)?"
(a) rectangle from 70 to 100, height 1/(100 − 70) = 1/30 (area = 30 × 1/30 = 1)
(b) (85 − 75)/30 = 10/30 = **0.3333**
(c) (80 − 70)/30 = 10/30 = **0.3333**

Q: "Friend is 0 to 30 min late, uniform. P(10 to 20 min late)? There is a 20% chance they arrive within the next ___ min." **A:** 10/30 = 0.3333; 0.20 × 30 = **6 min**. "Random number 0–10: P(exactly 4.35)?" **A:** 0.

Normal distribution

Normal = bell-shaped, symmetric, described by μ and σ. Properties: (1) symmetric about μ; (2) mean = median = mode = μ (highest point); (3) inflection points at μ ± σ; (4) total area = 1; (5) tails approach the axis but never touch; (6) Empirical Rule: 68% within 1σ, 95% within 2σ, 99.7% within 3σ; (7) area over an interval = proportion of the population in it = probability one random value is in it.

Standard normal Z = normal with μ = 0, σ = 1. Table V gives the area to the LEFT of z.

$z = \frac{x-\mu}{\sigma}$ (round z to 2 decimals) $x = \mu + z\cdot\sigma$
LEFT of z: read table · RIGHT of z: 1 − (left) · BETWEEN: (left of z_2) − (left of z_1)
Negative z: area left of −z = 1 − (area left of +z)

1. Draw the curve, mark μ, shade the region asked.
2. Convert each x to z (2 decimals).
3. Look up the area to the LEFT of each z (table below).
4. Left → done. Right → 1 − area. Between → subtract the two lefts.

Q: "(a) Area to the left of z = −1.10. (b) Area between z = −0.60 and z = 1.45. (c) X normal with μ = 80, σ = 10; find P(70 < X < 95)."
(a) table: **0.1357**
(b) left of 1.45 = 0.9265; left of −0.60 = 0.2743 → 0.9265 − 0.2743 = **0.6522**
(c) z = (70 − 80)/10 = −1.00; z = (95 − 80)/10 = 1.50 → 0.9332 − 0.1587 = **0.7745**

Q: "Wechsler IQ: μ = 100, σ = 15. (a) P(score above 125)? (b) P(below 90)? (c) Proportion between 110 and 140? (d) What score is the 98th percentile?"
(a) z = (125 − 100)/15 = 1.67 → left = 0.9525 → right = 1 − 0.9525 = **0.0475**
(b) z = (90 − 100)/15 = −0.67 → **0.2514**
(c) z = (110−100)/15 = 0.67 → 0.7486; z = (140−100)/15 = 2.67 → 0.9962 → 0.9962 − 0.7486 = **0.2476**
(d) area to the left = 0.98 → closest table value 0.9798 → z = 2.05 → x = 100 + 2.05(15) = **130.75 ≈ 131**

Q: "SAT math: μ = 516, σ = 116. P(less than 500)? P(more than 600)? P(between 525 and 550)?"
z = −0.14 → **0.4443** · z = 0.72 → 1 − 0.7642 = **0.2358** · z = 0.08 and 0.29 → 0.6141 − 0.5319 = **0.0822**

Find x from a percent / percentile ("backwards")

1. Draw & shade. Turn the question into an area to the LEFT (k-th percentile → 0.k; "top 10%" → left area 0.90; "middle 80%" → 0.10 and 0.90).
2. Find that area INSIDE the table (closest value); read z from the edges.
3. x = μ + z·σ.

Percentile	z	Percentile	z
5th	−1.645	95th	1.645
10th	−1.28	90th	1.28
20th	−0.84	80th	0.84
25th	−0.67	75th	0.67
2.5th	−1.96	97.5th	1.96
1st	−2.33	99th	2.33
2nd	−2.05	98th	2.05

Middle 80% ↔ z = ±1.28 · middle 90% ↔ ±1.645 · middle 95% ↔ ±1.96 · middle 99% ↔ ±2.575

Q: "Heights of 3-yr-old girls: μ = 38.72 in, σ = 3.17. (a) Height at the 20th percentile? (b) Heights that separate the middle 80%?"
(a) area 0.20 → z = −0.84 → x = 38.72 + (−0.84)(3.17) = **36.06 in**
(b) z = ±1.28 → 38.72 − 1.28(3.17) = **34.66** and 38.72 + 1.28(3.17) = **42.78 in**
Q: "P(shorter than 35 in)?" z = (35 − 38.72)/3.17 = −1.17 → **0.1210**

Interpreting an area; assessing normality

- "Area to the right of 200 is 0.2903" means: (1) the **proportion** of the population above 200 is 0.2903, and (2) the **probability** a randomly selected person is above 200 is 0.2903.
- **Normal?** Histogram: bell-shaped, one peak, roughly symmetric, few outliers. **Normal probability plot:** points close to a straight line → approximately normal.

Standard normal table — area to the LEFT of z (positive z; for negative z use 1 − value)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Reading the table: z = 1.67 → row **1.6**, column **.07** → 0.9525. Negative z: left of −1.67 = 1 − 0.9525 = 0.0475. **Backwards:** find the area inside the table (0.98 → 0.9798 is closest) → read z from the edges (row 2.0, column .05 → z = 2.05).

Which rule do I use? (Ch 4)

- One event, outcomes equally likely → N(E)/N(S). From data → frequency/total.
- "**OR**" → add; subtract the overlap unless disjoint.
- "**AND**" → multiply: P(E)·P(F) if independent, P(E)·P(F | E) if not.
- "**GIVEN** / of those who…" → conditional: cell ÷ given group.
- "**AT LEAST ONE**" → 1 − P(none). "**NOT**" → 1 − P(E).
- Counting: sequence → multiply; order matters → ${}_nP_r$; order doesn't → ${}_nC_r$.

Empirical Rule (bell-shaped only)

68% within μ ± 1σ · 95% within μ ± 2σ · 99.7% within μ ± 3σ
bands: 34% | 13.5% | 2.35% | 0.15% on each side of μ

Q: "IQ μ = 100, σ = 15. What % are between 85 and 130?" **A:** 85 = μ − 1σ, 130 = μ + 2σ → 34 + 34 + 13.5 = **81.5%**. (Exact z-table: 0.9772 − 0.1587 = 0.8185.)

DON'T LOSE POINTS

- "Or" → subtract the overlap once. "And" → the cell. "Given" → divide by the given group, not the grand total.
- Check Σ P(x) = 1 before calling a distribution valid.
- Sample vs. whole: binomial μ = np, σ = √(np(1−p)) — don't forget the √.
- "Fewer than 3" stops at 2; "at least 3" = 1 − P(≤ 2); "at most" includes the number.
- Uniform: probability = (length ÷ total width); never use z here.
- Normal: ALWAYS convert to z (2 decimals) first; table = area to the LEFT.
- "Above / more than" → 1 − left. "Between" → big area minus small area.
- Percentile question → area → z (from inside the table) → x = μ + zσ.
- Round probabilities to 4 decimals; show the formula every time.